

Q1

Apply the Dynamic programming algorithm and give the optimal solution to the matrix chain multiplication ABCD where the order for matrix.

$$A = 4 \times 5 \quad B = 5 \times 8 \quad C = 8 \times 3 \quad D = 3 \times 7$$

	A	B	C	D
A	0	160		
B		0	120	
C			0	
D				0

Minimum operations= _____

Best order of multiplication: _____

Q2

- The code length in Huffman codes depend on the frequency of occurrence of characters
☐ True ☐ False
- Greedy algorithm can solve all the dynamic programming problems.
☐ True ☐ False
- The time to run MERGESORT on a list of size n, $T(n) = T(n/2) + T(n/2) + T(\text{MERGE}(n/2))$
☐ True ☐ False
- In order to compare two algorithms, the same hardware and software environments must be used without any consideration of inputs type and its size
☐ True ☐ False
- Theoretical analysis allows us to evaluate the speed of an algorithm independent of the hardware/software environment.
☐ True ☐ False
- An algorithm which does NOT use the past results and uses them to find new results is:
 - Greedy Method
 - Divide and conquer
 - Dynamic programming
 - None of the mentioned

Q3

Using the dynamic programming approach, find the longest common subsequence between two strings:

X = ACBAED and Y = ABCADF.

Length _____

The subsequence: 1 _____ 2 _____

Q4

Insert integer keys A = {4, 13, 49, 6, 40, 33} in order into a hash table of size 9 using the hash function $h_1(k) = k \bmod 9$.

Collisions should be resolved via chaining, where collisions are stored at the end of a chain. Fill in the hash table after all keys have been inserted.

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Q5

A) Create a B-tree with order 5, min degree 2 the following item [24,52,90,40,11,86,61,50]

B) insert 29

C) delete 52,11